

Leo Shaw

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Education

Carnegie Mellon University

M.S. Artificial Intelligence Engineering

Pittsburgh, PA

Dec 2026

Rutgers University

B.S. Mechanical Engineering; B.A. Computer Science

New Brunswick, NJ

Dec 2024

Technical Skills

Languages: Python, C++, C, Java, MATLAB

AI & Data: PyTorch, TensorFlow, OpenCV, YOLO, LLM/SLM workflows, OCR/VLM pipelines, RAG, data pipelines

Robotics & Systems: Stable-Baselines3, PPO, CARLA, Gymnasium, Docker, PostgreSQL, Linux, Raspberry Pi, embedded systems, SOLIDWORKS

Projects

Graduate AI Research Lead

May 2026 - Present

Python, Docker, PostgreSQL, data pipelines, LLMs/SLMs, OCR/VLM, RAG

- Developed **multiple training-data pipelines** for a 6B-parameter manufacturing SLM, transforming technical documents, figures, tables, and process logs into structured inputs for retrieval, pretraining, and reasoning tasks.
- Built a Dockerized PostgreSQL backbone for a **3.5B-5B-token corpus** spanning RAG/pretraining data, supervised reasoning examples, tree-of-thought traces, and semiconductor process telemetry.
- Created **OCR/VLM pipelines** that produced section-aware 200-800-token samples with **source metadata**, **data lineage**, consistent schemas, and **quality gates** before database ingestion.
- Developed and evaluated retrieval, chain-of-thought, and **tree-of-thought workflows** for complex semiconductor manufacturing reasoning.

Autonomous Driving Planner with Reinforcement Learning

Jan 2026 - May 2026

Python, PyTorch, Stable-Baselines3, CARLA, PPO, A, PID*

- Built a **closed-loop CARLA planning stack** for highway merges and unprotected left turns, combining A* global routing, **PPO trajectory generation**, PID control, and route-relative observations.
- Developed the complete **training and evaluation pipeline**, including scenario generation, reward shaping, rollout analysis, policy benchmarking, and closed-loop testing.

Reinforcement Learning for Dynamic Obstacle Navigation

Jan 2026 - May 2026

Python, Gymnasium, Stable-Baselines3, PPO, A, NumPy, Matplotlib*

- Built a **custom simulation environment** for training **autonomous drone policies** in static and dynamic-obstacle scenarios.
- Trained PPO agents for obstacle-aware path planning and **benchmarked learned policies against A*** search under consistent navigation conditions.
- Designed **evaluation workflows** measuring task success, path efficiency, collision avoidance, and policy adaptability across varied obstacle layouts.

Real-Time Computer Vision and Robotic Sorting

Jan 2026 - May 2026

Python, YOLO, OpenCV, Raspberry Pi, inverse kinematics, embedded systems

- Developed a **real-time perception and robotic sorting pipeline** for bottles and cans using YOLO/OpenCV, homography-based camera calibration, inverse kinematics, and delta-arm motion control.
- Integrated **distributed perception and actuation** across a Raspberry Pi, cloud vision pipeline, servo-driven delta arm, and pneumatic end effector, achieving up to **99% classification accuracy** and approximately 50% end-to-end sorting accuracy while quantifying hardware and latency bottlenecks.

Experience

Field Service Engineer I

Santa Clara, CA / Ulvac Technologies, Oct 2024 - Aug 2025

PLC/HMI systems, valve-control software, diagnostics, system reliability, production debugging

- Delivered **production system upgrades** across PLC/HMI, RF, vacuum, and motion-handling subsystems, reducing downtime and improving process stability.
- Led **root-cause debugging** with engineering and manufacturing teams, standardizing troubleshooting procedures to accelerate fault diagnosis.
- Modified **valve-control software** and wiring logic to support upgraded tool configurations, integrate new subsystem hardware, and improve automated process control.